

- 6** A self-driving tractor sows seeds and harvests crops for a farmer. The self-driving tractor moves automatically through the field, turning at each end and stopping when it has finished.

- (a)** One reason the self-driving tractor is a robot is because it has a mechanical structure that includes wheels, an engine, and a framework.

Give **one** other reason why the self-driving tractor is an example of a robot.

.....
 [1]

- (b)** If a person is detected within 3 metres whilst the tractor is moving, the self-driving tractor must stop.

- (i)** Explain how an infra-red sensor, microprocessor and actuator can be used to stop the self-driving tractor if a person is detected within 3 metres.

.....

 [5]

- (ii)** Identify **one** other sensor that the self-driving tractor might use and how it will use the sensor.

Sensor

Use

..... [2]